ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Circular Motion

(English Medium)

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Kinematics of circular motion:

1. Suppose a disc is rotating counter clockwise in the plane of the paper then **CR0143**

(A) It's angular velocity vector will be perpendicular to the page pointing up out of the page

(B) It's angular velocity vector will be perpendicular to the page pointing in wards

(C) It's angular velocity vector acts along the tangent to the disc.

(D) None of the above is correct

2. A circular disc is rotating about its own axis at the rate of 200 revolutions per minute. Two particles P, Q of disc are at distances 5cm, 10cm from axis of rotation. The ratio of angular velocities of P and Q is:

(A) 1:2 (B) 1:1 (C) 2:1 (D) 4:1

CR0111

3. A circular disc is rotating about its own axis at uniform rate completes 30 rotations in one minute.The angular velocity of disc in rad s^{-1} is: **CR0144**

(A) 2π (B) π (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

4. A stationary wheel starts rotating about its own axis at uniform angular acceleration 8rad/s^2. The time taken by it to complete 77 rotations is: **CR0145**

(A) 5.5 sec (B) 7 sec (C) 11 sec (D) 14 sec

5. A stationary wheel starts rotating about its own axis at constant angular acceleration. If the wheel completes 50 rotations in first 2 seconds, then the number of rotations made by it in next two seconds is: **CR0146**

(A) 75 (B) 100 (C) 125 (D) 150

6. A particle is moving along a circular path with uniform speed. Through what angle does its angular velocity change when it completes half of the circular path ?

(A) 0° (B) 45° (C)180° (D) 360°

CR0112

7. If angular velocity of a disc depends an angle rotated θ as $\omega = \theta^2 + 2\theta$, then its angular acceleration α at $\theta = 1$ rad is :

(A) 8 rad/s^2 (B) 10rad/s^2 (C) 12 rad/s^2 (D) None of these

CR0113

8. A body is executing circular motion in the vertical plane containing directions. If the direction of velocity $(\vec{v})$ at the top most point is towards west, what is the direction of angular velocity $(\vec{\omega})$?

(A) east (B) west (C) north (D) south

CR0030

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9. If the magnitude of velocity in the previous question is decreasing with time, what is the direction of angular acceleration $(\vec{\alpha})$?

(A) east (B) west (C) north (D) south

CR0031

10. A circular plate is rotating about its own axis at an angular velocity 100 revolutions per minute. The linear velocity of a particle P of plate at a distance 4.2 cm from axis of rotation is:

(A) 0.22 m/s (B) 0.44 m/s (C) 2.2 m/s (D) 4.4 m/s **CR0147**

11. The second's hand of a watch has length 6 cm. Speed of end point and magnitude of difference of velocities at two perpendicular positions will be

(A) 2π & 0 mm/s (B) $2\sqrt{2}\ \pi$ & 4.44 mm/s

(C) $2\sqrt{2}\ \pi$ & 2π mm/s (D) 2π & $2\sqrt{2}\ \pi$ mm/s

CR0033

12. A particle is moving at uniform speed 2 ms^{-1} along a circle of radius 0.5m. The centripetal acceleration of particle is:

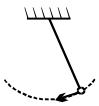
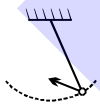
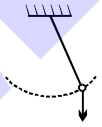
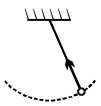
(A) $1ms^{-2}$ (B) $2ms^{-2}$ (C) $4ms^{-2}$ (D) $8ms^{-2}$

CR0114

13. A point size body is moving along a circle at an angular velocity 2.8 $rads^{-1}$. If centripetal acceleration of body is $7ms^{-2}$ then its speed is :

(A) 1.25 ms^{-1} (B) 2.5 ms^{-1} (C) 3.5 ms^{-1} (D) 7 ms^{-1} **CR0148**

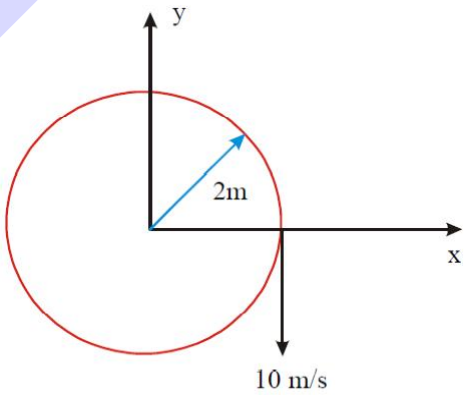
14. Which vector in the figures best represents the acceleration of a pendulum mass at the intermediate point in its swing?

(A)  (B)  (C)  (D) 

CR0039

15. A particle is moving along a circular path in xy-plane. When it crosses x-axis, it has an acceleration along the path of 1.5 m/s^2 and is moving with a speed of 10 m/s in -ve y - direction. The total acceleration is:

CR0149



(A) $50\hat{i}-1.5\hat{j}\ m/s^2$ (B) $10\hat{i}-1.5\hat{j}\ m/s^2$ (C) $-50\hat{i}-1.5\hat{j}\ m/s^2$ (D) $-1.5\hat{i}-50\ \hat{j}\ m/s^2$

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16. A cyclist riding with a speed of 27kmph. As he approaches a circular turn on the road of radius 80m, he applies brakes and reduces his speed at the constant rate of 0.50 m/s every second. The net acceleration of cyclist on the circular turn is:

(A) $0.5 m/s^2$ (B) $0.87 m/s^2$ (C) $0.56 m/s^2$ (D) $1 m/s^2$

CR0115

17. A particle moves in a circle of radius 20 cm. Its linear speed is given by v = 2t where t is in s and v in m/s. Then

a) the radial acceleration at t = 2s is $80 ms^{-2}$.

b) tangential acceleration at t = 2s is $2 ms^{-2}$.

c) net acceleration at t = 2 s is greater than $80 ms^{-2}$.

d) tangential acceleration remains constant in magnitude.

(A) only a,b,c are correct (B) only a,b,d are correct

(C) only a,c,d are correct (D) all a,b,c,d are correct **CR0116**

18. If the radii of circular path of two particles are in the ratio of 1 : 2, then in order to have same centripetal acceleration, their speeds should be in the ratio of : **CR0150**

(A) 1 : 4 (B) 4 : 1 (C) $1:\sqrt{2}$ (D) $\sqrt{2}:1$

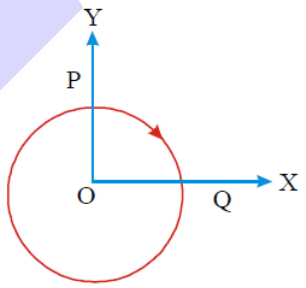
19. A particle is kept fixed on a turntable rotating uniformly. As seen from the ground, the particle goes in a circle, its speed is 20 cm/s and acceleration is $20 cm/s^2$. The particle is now shifted to a new position to make the radius half of the original value. The new values of the speed and acceleration will be:

(A) 10 cm/s, $10 cm/s^2$ (B) 10 cm/s, $80 cm/s^2$

(C) 40 cm/s, $10 cm/s^2$ (D) 40 cm/s, $40 cm/s^2$

CR0117

20. A particle moves in a circle of radius 4cm clockwise at constant speed 2cm/sec. If $\hat{x}$ and $\hat{y}$ are unit acceleration vectors along x and y-axes respectively (in cm/sec^2), the acceleration of the particle at the instant halfway between P and Q is given by: **CR0151**



(A) $-4(\hat{x}+\hat{y})$ (B) $4(\hat{x}+\hat{y})$ (C) $-(\hat{x}+\hat{y})\frac{1}{\sqrt{2}}$ (D) $(\hat{x}-\hat{y})4$

21. A particle moves on a circle of radius r with centripetal acceleration as function of time as $a_c = k^2rt^2$ where k is a positive constant. Find the resultant acceleration.

(A) kt^2 (B) kr (C) $kr\sqrt{k^2t^4+1}$ (D) $kr\sqrt{k^2t^2-1}$

CR0118

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22. A particle P is moving in a circle of radius 'a ' with a uniform speed v . C is the centre of the circle and AB is a diameter. when passing through B the angular velocity of P about A and C are in the ratio:

(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 1 : 3

CR0119

23. A ball is projected with $20\sqrt{2}$ m/s at angle 45° with horizontal. The angular velocity of the particle at highest point of its journey about point of projection is :

(A) 0.1 rad/s (B) 0.2 rad/s (C) 0.3 rad/s (D) 0.4 rad/s

CR0120

24. A stone is thrown horizontally with a velocity of 10m/sec. Find the radius of curvature of it's trajectory at the end of 3 s after motion began. ($g = 10m/s^2$)

(A) $10\sqrt{10}$m (B) $100\sqrt{10}$m (C) $\sqrt{10}$m (D) 100m

CR0121

Kinematics of circular motion:

25. A particle is projected with a speed u at an angle θ with the horizontal. Consider a small part of its path near the highest position and take it approximately to be a circular arc. What is the radius of this circle?

(A) $\frac{u^2 \sin^2\theta}{g}$ (B) $\frac{u^2 \cos^2\theta}{g}$ (C) $\frac{u^2 \tan^2\theta}{g}$ (D) $\frac{u^2}{g}$

CR0122

26. A particle of mass m is tied to a light string and rotated with a speed v along a circular path of radius r. If T = tension in the string and mg = gravitational force on the particle then the actual forces acting on the particle are :

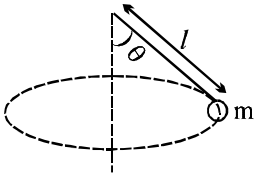
(A) mg and T only

(B) mg, T and an additional force of $\frac{mv^2}{r}$ directed inwards.

(C) mg, T and an additional force of $\frac{mv^2}{r}$ directed outwards.

(D) only a force $\frac{mv^2}{r}$ directed outwards. **CR0038**

27. A conical pendulum is moving in a circle with angular velocity ω as shown. If tension in the string is T, which of following equations are correct ?

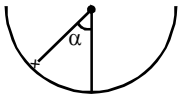


(A) $T = m\omega^2 l$ (B) $T\sin\theta = m\omega^2 l$ (C) $T = mg\cos\theta$ (D) $T = m\omega^2 l \sin\theta$

CR0040

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28. A insect crawls up a hemispherical surface very slowly (see the figure). The coefficient of friction between the surface and the insect is 1/3. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value of α is given **[IIT-JEE 2001]**



(A) $\cot\alpha = 3$ (B) $\tan\alpha = 3$ (C) $\sec\alpha = 3$ (D) $\operatorname{cosec}\alpha = 3$

CR0043

29. A vehicle can travel round a curve at a higher speed when the road is banked than when the road is level. This is because

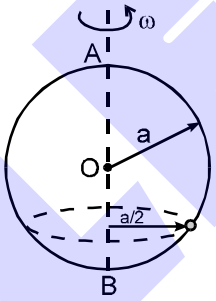
(A) banking increases the coefficient of friction

(B) banking increases the radius,

(C) the normal reaction has a horizontal component,

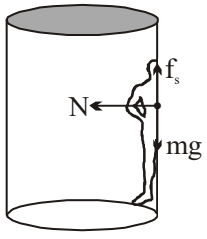
(D) when the track is banked the weight of the car acts down the incline. **CR0123**

30. A smooth wire is bent into a vertical circle of radius ***a***. A bead P can slide smoothly on the wire. The circle is rotated about vertical diameter AB as axis with a constant speed ω as shown in figure. The bead P is at rest w.r.t. the wire in the position shown. Then ω^2 is equal to : **CR0152**



(A) $\frac{2g}{a}$ (B) $\frac{2g}{a\sqrt{3}}$ (C) $\frac{g\sqrt{3}}{a}$ (D) $\frac{2a}{g\sqrt{3}}$

31. In a rotor, a hollow vertical cyclindrical structure rotates about its axis and a person rests against the inner wall. At a particular speed of the rotor, the floor below the person is removed and the person hangs resting against the wall without any floor. If athe radius of the rotor is 2m and the coefficient of static friction between the wall and the person is 0.2, find the minimum speed at which the floor may be removed. Take $g = 10\ m/s^2$.

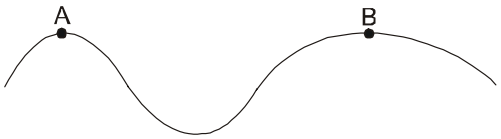


(A) 10 m/s (B) 20 m/s (C) 50 m/s (D) 100 m/s

CR0124

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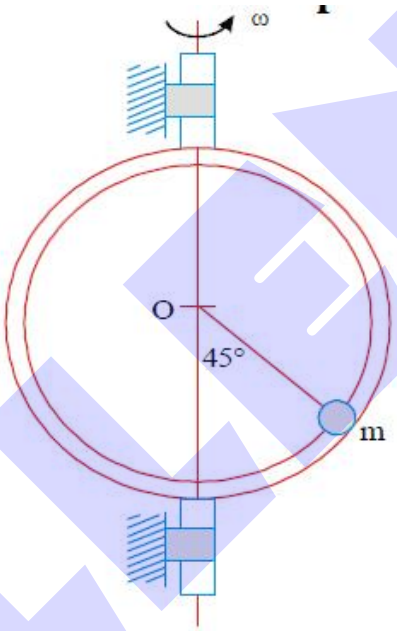
32. A car moves at a constant speed on a road as shown in figure. The normal force by the road on the car is N_A and N_B when when it is at the points A and B respectively.



(A) $N_A = N_B$ (B) $N_A > N_B$ (C) $N_A < N_B$ (D) insufficient information

CR0125

33. A small bead of mass m is carried by a circular hoop having centre at O and radius $\sqrt{2}m$ which rotates about a fixed vertical axis. The coefficient of friction between beed and hoop is $\mu = 0.5$. The maximum angular speed of the hoop for which the bead does not have relative motion with respect to hoop.($g = 10m/s^2$) **CR0153**



(A) $\sqrt{5}$ (B) $\sqrt{10}$ (C) $\sqrt{15}$ (D) $\sqrt{30}$

MULTIPLE CORRECT TYPE QUESTIONS

Kinematics of circular motion:

34. A car runs around a curve of radius 10 m at a constant speed of 10 ms^{-1}. Consider the time interval for which car covers a curve of 120° arc :-

(A) Resultant change in velocity of car is $10\sqrt{3}$ ms^{-1}

(B) Instantaneous acceleration of car is 10 ms^{-2}

(C) Average acceleration of car is $\frac{5}{24}$ ms^{-2}

(D) Instantaneous and average acceleration are same for the given period of motion.

CR0046

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35. A car is moving with constant speed on a rough banked road.

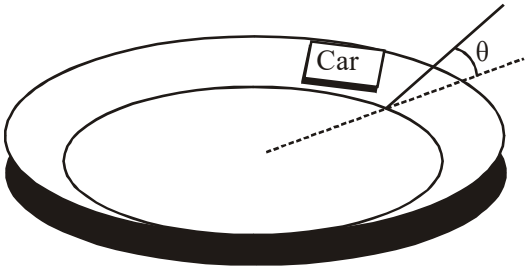
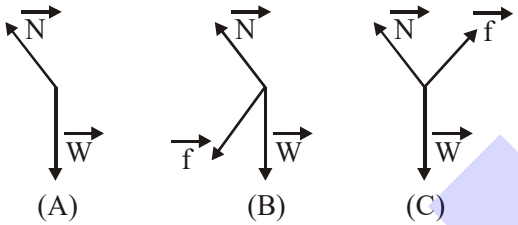


Figure shows the free body diagram of car in three situation A, B & C respectively:-



(A) Car in A has more speed than car in C
(B) Car in A has less speed than car in B
(C) FBD for car in A is not possible
(D) If $\mu > \tan\theta$ the FBD for car C is not possible **CR0047**

36. A body moves on a horizontal circular road of radius r, with a tangential acceleration a_T. Coefficient of friction between the body and road surface is μ. It begin to slip when it's speed is v, then:

(A) $v^2 = \mu rg$

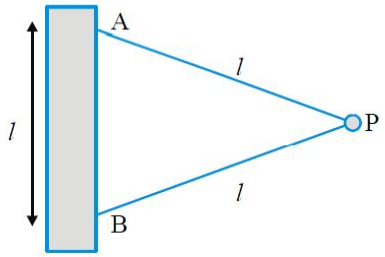
(B) $\mu g = \frac{v^2}{r} + a_T$

(C) $\mu^2 g^2 = \frac{v^4}{r^2} + a_T^2$

(D) The force of friction makes an angle $\tan^{-1}\left(\frac{v^2}{a_T \times r}\right)$ with direction of motion at point of slipping.

CR0127

37. A particle P of mass m is attached to a vertical axis by two strings AP and BP of length ℓ each. The separation AB = ℓ, P rotates around the axis with an angular velocity ω. The tensions in the two strings are T_1 and T_2. Then



(A) $T_1 = T_2$

(B) $T_1 + T_2 = m\omega^2\ell$

(C) $T_1 - T_2 = 2mg$

(D) BP will remain taut only if $\omega \geq \sqrt{\frac{2g}{\ell}}$.

CR0128

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38. A heavy particle is tied to the end A of a string of length 1.6 m. Its other end O is fixed. It revolves as a conical pendulum with the string making 60° with the vertical. Then

(A) its period of revolution is $\frac{4\pi}{7}$ sec.

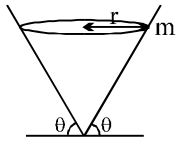
(B) the tension in the string is double the weight of the particle

(C) the velocity of the particle = $2.8\sqrt{3}$ m/s

(D) the centripetal acceleration of the particle is $9.8\sqrt{3}$ m/s^2. **CR0048**

39. A ball of mass 'm' is rotating in a circle of radius 'r' with speed v inside a smooth cone as shown in figure. Let N be the normal reaction on the ball by the cone, then choose the correct option.

(A) $N = mg\cos\theta$ (B) $g\sin\theta = \frac{v^2}{r}\cos\theta$

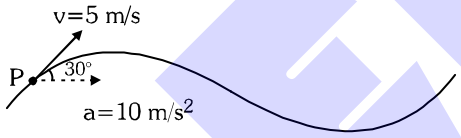


(C) $N\sin\theta - \frac{mv^2}{r} = 0$ (D) none of these

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 40 and 41

A particle is moving on given curve path. At certain instant its velocity and acceleration shown in figure



40. Rate of change of speed of particle at P is

(A) $5\sqrt{3}\,m/s^2$ (B) 10 m/s^2 (C) 5 m/s^2 (D) None of these

CR0129

41. Radius of curvature at point P is :

(A) $5\sqrt{3}$ m (B) 10 m (C) 5 m (D) None of these

CR0130

Paragraph for Question Nos. 42 to 44

Two blocks of mass m_1 =10kg and m_2 = 5kg, connected to each other by a massless inextensible string of length 0.3m are placed along a diameter of a turn table. The coefficient of friction between the table and m_1 is 0.5 while there is no friction between m_2 and the table. The table is rotating with an angular velocity of 10rad/s about a vertical axis passing through its centre O. The masses are placed along the diameter of the table on either side of the centre O such that the mass m_1 is at a distance 0.124m from O. The masses are observed to be at rest with respect to an observer on the turn table.

42. Calculate the fricitonal force on m_1.

(A) 28 N (B) 32 N (C) 36 N (D) 40 N

CR0131

43. What should be the minimum angular speed of the turn table so that the masses will slip from this position?

(A) 12.82 rad/s (B) 10.28 rad/s (C) 13.56 rad/s (D) 11.67 rad/s

CR0132

44. How should the masses m_2 be placed with the string remaining taut, so that there is no frictional force acting on the mass m_1 ?

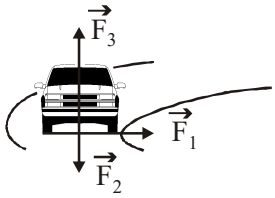
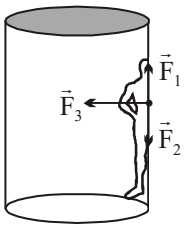
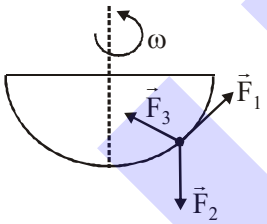
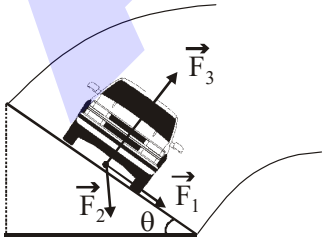
(A) 0.2 m (B) 0.3 m (C) 0.4 m (D) 0.5 m

CR0133

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MATRIX MATCH TYPE QUESTION

45. Column-I shows certain situations and column-2 shows information about forces.

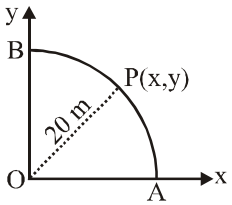
Column - I	**Column - II**
(A) Situation  Front view of a car rounding a curve with constant speed.	(P) $\vec{F}_1+\vec{F}_2+\vec{F}_3$ is centripetal force.
(B)  Passengers in a rotor not sliding relative to rotor wall cylindrical rotor is rotating with constant angular velocity about its symmetry axis.	(Q) $\vec{F}_1$ is static friction.
(C)  Particle kept on rough surface of a bowl, no relative motion of particle in bowl, bowl has constant angular velocity	(R) $\vec{F}_1$ can be in direction opposite to that shown in figure.
(D)  Car moving on a banked road with constant speed, no sideways skidding	(S) $\vec{F}_1+\vec{F}_2=\vec{0}$
	(T) $\vec{F}_1+\vec{F}_2+\vec{F}_3=\vec{0}$

CR0068

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EXERCISE (O-2)

1. A point P moves in counter clockwise direction on a circular path as shown in the figure. The movement of 'P' is such that it sweeps out a length $s = t^2 + 5$, where s is in metres and t is in seconds. The radius of the path is 20 m. The acceleration of 'P' when $t = 5\sqrt{\frac{3}{10}}$ seconds is nearly :



(A) 2 m/s^2 (B) 1.5 m/s^2 (C) 2.5 m/s^2 (D) 3 m/s^2

CR0034

2. The magnitude of displacement of a particle moving in a circle of radius a with constant angular speed ω varies with time t as :-

(A) 2 a sinωt (B) $2a \sin\frac{\omega t}{2}$ (C) 2a cos ωt (D) $2a \cos\frac{\omega t}{2}$

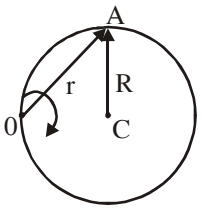
CR0053

3. A particle moves in a circular path such that its speed v varies with distance as $v = \alpha\sqrt{s}$ where α is a positive constant. Find the acceleration of particle after traversing a distance S?

(A) $\alpha^2\sqrt{\frac{1}{4}-\frac{S^2}{R^2}}$ (B) $\alpha^2\sqrt{\frac{1}{4}+\frac{S^2}{R^2}}$ (C) $\alpha\sqrt{\frac{1}{2}+\frac{S^2}{R^2}}$ (D) $\alpha^2\sqrt{\frac{1}{2}+\frac{S^2}{R^2}}$

CR0134

4. A particle A moves along a circle of radius R=50 cm so that its radius vector r relative to the point O (figure) rotates with the constant angular velocity ω=0.40 rad/s. Then modulus of the velocity of the particle, and the modulus of its total acceleration will be



(A) v= 0.4 m/s, a = 0.4 m/s^2 (B) v = 0.32 m/s, a = 0.32 m/s^2
(C) v = 0.32 m/s, a = 0.4 m/s^2 (D) v = 0.4 m/s, a = 0.32 m/s^2

CR0057

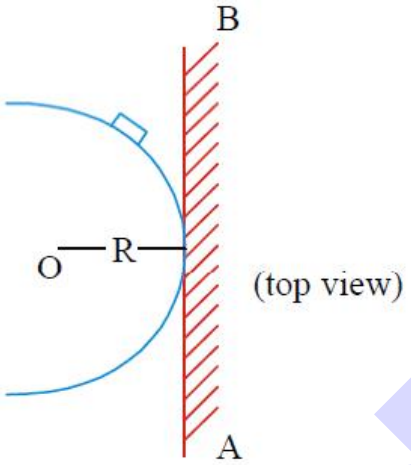
5. A particle is moving in a circular path. The acceleration and momentum of the particle at a certain moment are $\vec{a} = \left(4\hat{i} + 3\hat{j}\right)$ m/s^2 and $\vec{p} = \left(8\hat{i} - 6\hat{j}\right)$ kg–m/s. The motion of the particle is

(A) uniform circular motion (B) accelerated circular motion
(C) de–accelerated circular motion (D) we can not say anything with $\vec{a}$ and $\vec{p}$ only

CR0056

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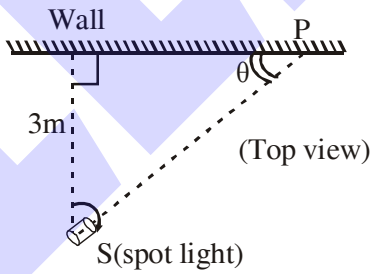
6. As shown in figure 'AB' represents an infinite wall tangential to a horizontal semi circular track. 'O' is a point source of light on the ground at the centre of the circle. A block moves along the circular track with speed 'v' starting from the point where the wall touches the circle. If the velocity and acceleration of shadow along the length of the wall is respectively 'V' and 'a' then; **CR0154**



(A) $V = v\cos\frac{vt}{R}$ (B) $V = v\sec^2\left(\frac{vt}{R}\right)$

(C) $a = \frac{v^2}{R}\sec^2\left(\frac{vt}{R}\right)\tan\left(\frac{vt}{R}\right)$ (D) $a = \frac{2v^2}{R}\sec^2\left(\frac{vt}{R}\right)\tan\left(\frac{vt}{R}\right)$

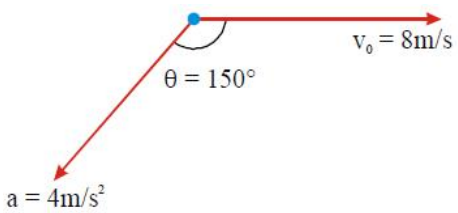
7. A spot light S rotates in a horizontal plane with a constant angular velocity of 0.1 rad/s. The spot of light P moves along the wall at a distance 3m. What is the velocity of the spot P when $\theta = 45°$?



(A) 0.6 m/s (B) 0.5 m/s (C) 0.4 m/s (D) 0.3 m/s

CR0035

8. The figure shows the velocity and acceleration of a point like body at the initial moment of its motion. The acceleration vector of the body remains constant. The minimum radius of curvature of trajectory of the body is: **CR0155**



(A) 2 meter (B) 4 meter (C) 8 meter (D) 16 meter

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9. A long horizontal rod has a bead which can slide along its length and is initially placed at a distance L from one end A of the rod. The rod is set in angular motion about A with a constant angular acceleration, α. If the coefficient of friction between the rod and bead is μ, and gravity is neglected, then the time after which the bead starts slipping is :– **[IIT-JEE 2000]**

(A) $\sqrt{\frac{\mu}{\alpha}}$ (B) $\frac{\mu}{\sqrt{\alpha}}$ (C) $\frac{1}{\sqrt{\mu\alpha}}$ (D) infinitesimal

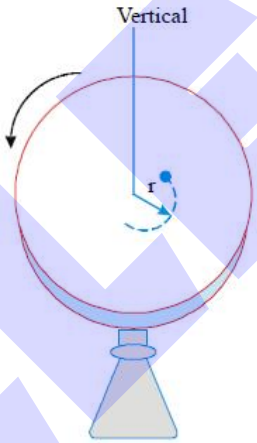
CR0042

10. If equation of path of moving particle is given by $y = 4x^2$, radius of curvature of path of particle at origin is :

(A) $\frac{1}{4}$m (B) $\frac{1}{2}$m (C) $\frac{1}{8}$m (D) None of these

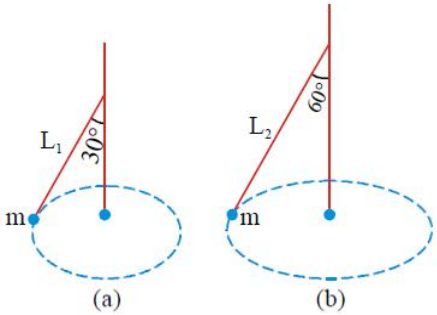
CR0135

11. A small coin of mass 80g is placed on the horizontal surface of a rotating disc. The disc starts from rest and is given a constant angular acceleration $\alpha = 2\text{rad/s}^2$. The coefficient of static friction between the coin and the disc is $\mu_s = 3/4$ and coefficient of kinetic friction is $\mu_k = 0.5$. The coin is placed at a distance r =1m from the centre of the disc. The magnitude of the resultant force on the coin exerted by the disc just before it starts slipping on the disc is: **CR0156**



(A) 0.2 N (B) 0.3 N (C) 0.4 N (D) 1N

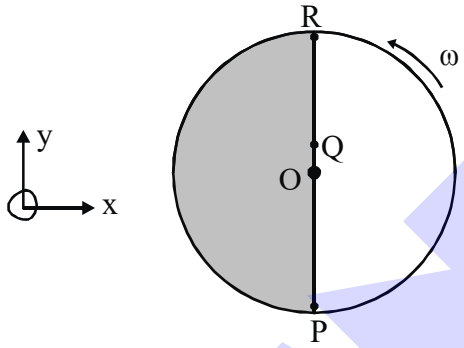
12. Two particles tied to different strings are whirled in a horizontal circle as shown in figure. The ratio of lengths of the strings so that they complete their circular path with equal time periods is: **CR0157**



(A) $\sqrt{\frac{3}{2}}$ (B) $\sqrt{\frac{1}{3}}$ (C) 1 (D) $\sqrt{3}$

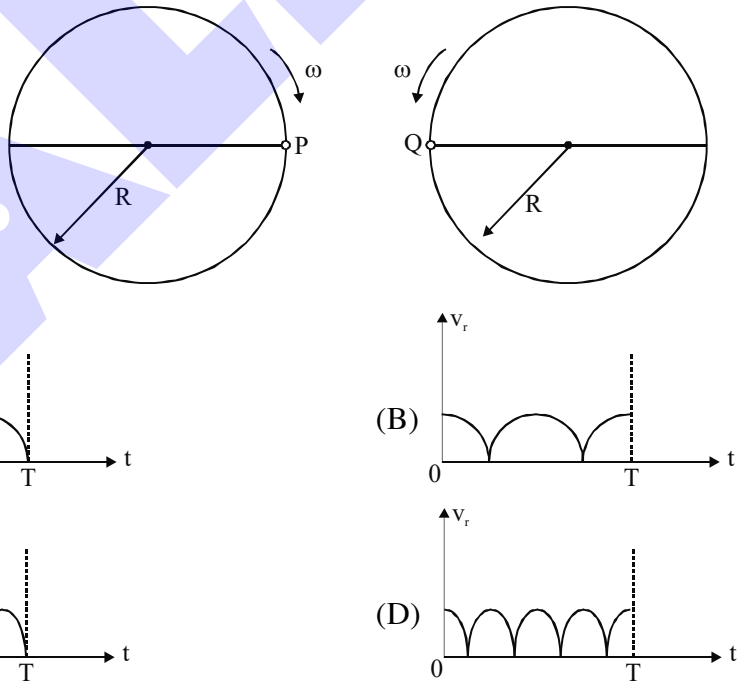
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13. Consider a disc rotating in the horizontal plane with a constant angular speed ω about its centre O. The disc has a shaded region on one side of the diameter and an unshaded region on the other side as shown in the figure. When the disc is in the orientation as shown, two pebbles P and Q are simultaneously projected at an angle towards R. The velocity of projection is in the y-z plane and is same for both pebbles with respect to the disc. Assume that (i) they land back on the disc before the disc has completed $\frac{1}{8}$ rotation, (ii) their range is less than half the disc radius, and (iii) ω remains constant throughout. Then **[IIT-JEE-2012]**



(A) P lands in the shaded region and Q in the unshaded region.
(B) P lands in the unshaded region and Q in the shaded region.
(C) Both P and Q land in the unshaded region.
(D) Both P and Q land in the shaded region. **CR0075**

14. Two identical discs of same radius R are rotating about their axes in opposite directions with the same constant angular speed ω. The discs are in the same horizontal plane. At time t =0, the points P and Q are facing each other as shown in the figure. The relative speed between the two points P and Q is v_r. In one time period (T) of rotation of the discs, v_r as a function of time is best represented by **[IIT-JEE 2012]**



CR0076

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EXERCISE (S)

Kinematics of circular motion:

1. A stone tied to the end of a string 80 cm long is whirled in a horizontal circle with a constant speed. If the stone makes 14 revolutions in 25 s, what is the magnitude and direction of acceleration of the stone ?

CR0001

2. A particle is revolving in a circle of radius 1m with an angular speed of 12 rad/s. At t = 0, it was subjected to a constant angular acceleration α and its angular speed increased to $(480/\pi)$ rotation per minute (rpm) in 2 sec. Particle then continues to move with attained speed. Calculate
(i) angular acceleration of the particle,
(ii) tangential velocity of the particle as a function of time.
(iii) acceleration of the particle at t = 0.5 second and at t = 3 second
(iv) angular displacement at t = 3 second.

CR0002

3. A particle is travelling in a circular path of radius 4m. At a certain instant the particle is moving at 20m/s and its acceleration is at an angle of 37° from the direction to the centre of the circle as seen from the particle
(i) At what rate is the speed of the particle increasing?
(ii) What is the magnitude of the acceleration? **CR0004**

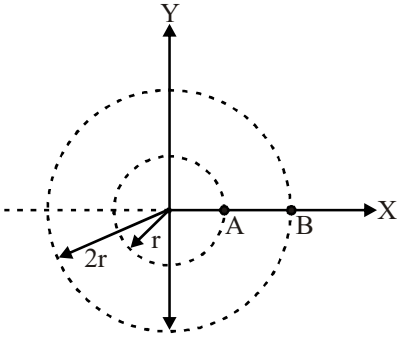
4. The angular displacement of a particle is given by $\theta = \omega_o t + \frac{1}{2}\alpha t^2$, where ω_o and α are constant and $\omega_o = 1$ rad/s, $\alpha = 1.5$ rad/s^2. Find the angular velocity at time t = 2s. **CR0158**

5. A disc starts from rest and on the application of a torque, it gains an angular acceleration given by $\alpha = 3t - t^2$. Calculate the angular velocity after 2 s. **CR0159**

6. A disk rotates about its central axis starting from rest and accelerates with constant angular acceleration. At one instant it is rotating at 12 rad/s and after 80 radian of more angular displacement, its angular speed becomes 28 rad/s. How much time (seconds) does the disk takes to complete the mentioned angular displacement of 80 radians.

CR0005

7. Two particles A and B are moving in a horizontal plane anticlockwise on two different concentric circles with different constant angular velocities 2ω and ω respectively. Find the relative velocity (in m/s) of B w.r.t. A after time $t = \pi/\omega$. They both start at the position as shown in figure (Take ω = 3rad/sec, r = 2m)



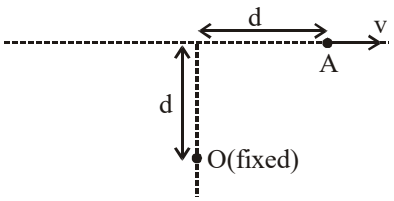
CR0006

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8. A particle moves in the x-y plane with the velocity $\vec{v} = a\hat{i} + bt\hat{j}$. At the instant $t = a\sqrt{3}/b$ the magnitude of tangential, normal and total acceleration are _____, _____, & _____.

CR0009

9. Find angular velocity of A with respect to O at the instant shown in the figure.



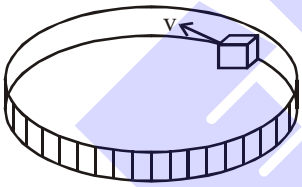
CR0007

10. A stone is thrown horizontally with the velocity 15m/s. Determine the tangential and normal accelerations of the stone in 1 second after it begins to move.

CR0008

Dynamics of circular motion

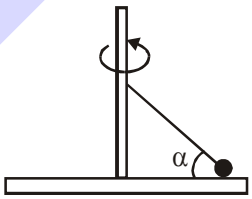
11. A block of mass m moves with speed v against a smooth, fixed vertical circular groove of radius r kept on smooth horizontal surface.



Find :
(i) normal reaction of the floor on the block.
(ii) normal reaction of the vertical wall on the block.

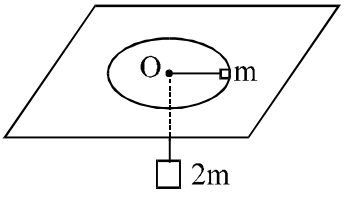
CR0011

12. A circular platform rotates around a vertical axis with angular velocity $\omega = 10$ rad/s. On the platform is a ball of mass 1 kg, attached to the long axis of the platform by a thin rod of length 10 cm ($\alpha = 30°$). Find normal force exerted by the ball on the platform (in newton). Friction is absent.



CR0016

13. A mass m rotating freely in a horizontal circle of radius 1 m on a frictionless smooth table supports a stationary mass 2m, attached to the other end of the string passing through smooth hole O in table, hanging vertically. Find the angular velocity of rotation.



CR0014

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14. A stone of mass 0.25 kg tied to the end of a string is whirled round in a circle of radius 1.5 m with a speed of 40 rev/min in a horizontal plane. What is the tension in the string ? What is the maximum speed with which the stone can be whirled around if the string can withstand a maximum tension of 200 N ?

CR0013

15. An aircraft executes a horizontal loop at a speed of 720 km/h with its wings banked at 15°. What is the radius of the loop ?

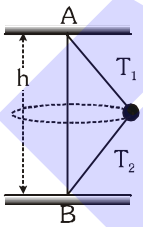
CR0017

16. A block of mass m = 20 kg is kept at a distance R = 1m from central axis of rotation of a round turn table (A table whose surface can rotate about central axis). Table starts from rest and rotates with constant angular acceleration, $\alpha = 3$ rad/sec^2. The friction coefficient between block and table is $\mu = 0.5$.

At time t = $\frac{x}{3}$ sec from starting of motion (i.e. t = 0 sec) the block is just about to slip. Find the value of x.

CR0018

17. A particle is attached by means of two equal strings to points A and B in the same vertical line and decribe a horizontal circle with a uniform angular speed. If the angular speed of the particle is $2\sqrt{(2g/h)}$ with AB = h, Find the ratio of the tension of the string.

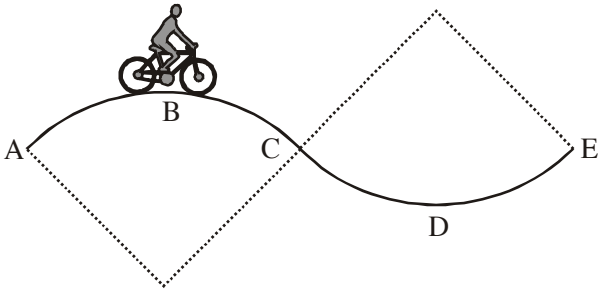


CR0136

18. A turn of radius 20m is banked for the vehicles going at a speed of 36 km/h. If the coeffient of static friction between the road and the tyre is 0.4, what are the possible speeds of a vehicle so that it neither slips down nor skids up?

CR0137

19. A track consists of two circular parts ABC and CDE of equal radius 100m and joined smoothly as shown in figure. Each part subtends a right angle at its centre. A cycle weighing 100 kg together with the rider travels at a constant speed of 18 km/hr on the track. (a) Find the normal contact force by the road on the cycle when it is at B and at D. (b) Find the force of friction exerted by the track on the tyres when the cycle is at B, C and D. (c) Find the normal force between the road and the cycle just before and just after the cycle crosses C. (d) What should be the minimum friction coefficient between the road and the tyre, which will ensure that the cyclist can move with constant speed? Take g = 10 m/s^2.



CR0138

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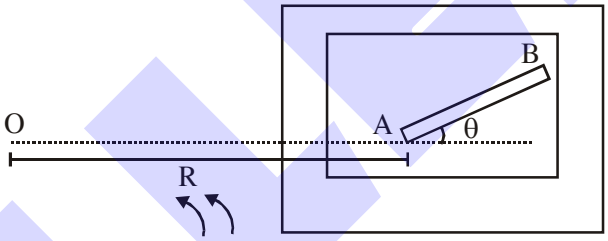
20. A hemispherical bowl of radius R is rotated about its axis of symmetry which is kept vertical. A small block is kept in the bowl at a position where the radius makes an angle θ with the vertical. The block rotates with the bowl without any slipping. The friction coefficient between the block and the bowl surface is μ. Find the range of the angular speed for which the block will not slip.

CR0139

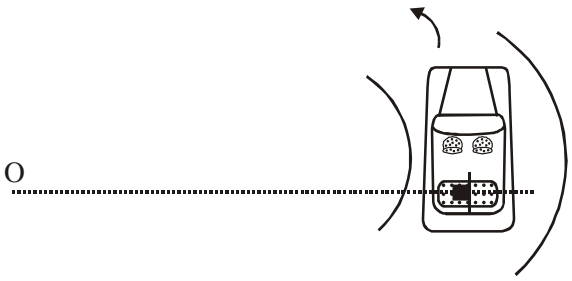
21. A block of mass m moves on a horizontal circle against thw wall of a cylindrical room of radius R. The floor of the room on which the block moves is smooth but the friction coefficient between the wall and the block is μ. The block is given and initial speed v_0. As a function of the speed v write (a) the normal force by the wall on the block, (b) the frictional force by the wall and (c) the tangential acceleration of the block. (d) Integrate the tangential acceleration $\left(\frac{dv}{dt} = v\frac{dv}{ds}\right)$ to obtain the speed of the block after one revolution.

CR0140

22. A table with smooth horizontal surface is fixed in a cabin that rotates with a uniform angular velocity ω in circular path of radius R (in figure). A smooth groove AB of length L(<<R) is made on the surface of the table. The groove makes an angle θ with the radius OA of the circle in which the cabin rotates. A small particle is kept at the point A in the groove and is released to move along AB. Find the time taken by the particle to reach the point B. **CR0160**



23. A car moving at a speed of 36 km/hr is taking a turn on a circular road of radius 50m. A small wooden plate is kept on the seat with its plane perpendicular to the radius of the circular road in figure. A small block of mass 100g is kept on hte seat which rests against the plate. The friction coefficient between the block and the plate is μ = 0.58. (a) Find the normal contact force exerted by the plate on the block. (b) The plate is slowly turned so that the angle between the normal to the plate and the radius of the road slowly increases. Find the angle at which the block will just start sliding on the plate. **CR0161**



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EXERCISE (JM)

1. Concrete mixture is made by mixing cement, stone and sand in a rotating cylindrical drum. If the drum rotates too fast, the ingredients remain stuck to the wall of the drum and proper mixing of ingredients does not take place. The maximum rotational speed of the drum in revolutions per minute(rpm) to ensure proper mixing is close to : (Take the radius of the drum to be 1.25 m and its axle to be horizontal): **[JEE Main (Online) - 2016]**

(1) 8.0 (2) 0.4 (3) 1.3 (4) 27.0

CR0072

2. A particle is moving with a uniform speed in a circular orbit of radius R in a central force inversely proportional to the nth power of R. If the period of rotation of the particle is T, then, **[JEE Main-2018]**

(1) $T \propto R^{\frac{n}{2}+1}$ (2) $T \propto R^{(n+1)/2}$ (3) $T \propto R^{n/2}$ (4) $T \propto R^{3/2}$ for any n

CR0073

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

3. A body is projected at t = 0 with a velocity 10 ms^{-1} at an angle of 60° with the horizontal. The radius of curvature of its trajectory at t = 1s is R. Neglecting air resistance and taking acceleration due to gravity g = 10 ms^{-2}, the value of R is : **[JEE Main-2019_Jan]**

(1) 2.5 m (2) 10.3 m (3) 2.8 m (4) 5.1 m

CR0100

4. A particle is moving along a circular path with a constant speed of 10 ms^{-1}. What is the magnitude of the change is velocity of the particle, when it moves through an angle of 60° around the centre of the circle? **[JEE Main-2019_Jan]**

(1) zero (2) 10 m/s (3) $10\sqrt{3}$ m/s (4) $10\sqrt{2}$ m/s

CR0101

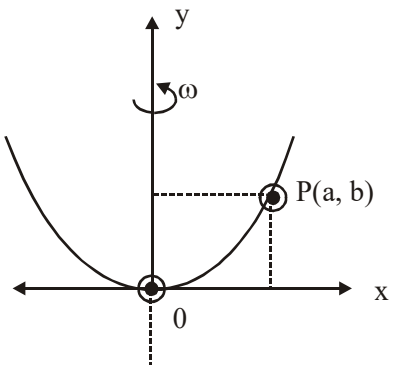
5. A spring mass system (mass m, spring constant k and natural length *l*) rest in equilibrium on a horizontal disc. The free end of the spring is fixed at the centre of the disc. If the disc together with spring mass system, rotates about it's axis with an angular velocity ω, ($k >> m\omega^2$) the relative change in the length of the spring is best given by the option : **[JEE Main-2020_Jan]**

(1) $\frac{2m\omega^2}{k}$ (2) $\frac{m\omega^2}{3k}$ (3) $\sqrt{\frac{2}{3}}\left(\frac{m\omega^2}{k}\right)$ (4) $\frac{m\omega^2}{k}$

CR0102

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6. A bead of mass m stays at point P(a, b) on a wire bent in the shape of a parabola $y = 4Cx^2$ and rotating with angular speed ω (see figure). The value of ω is (neglect friction) : **[JEE Main-2020_Sep.]**



(1) $\sqrt{\frac{2gC}{ab}}$ (2) $2\sqrt{2gC}$ (3) $\sqrt{\frac{2g}{C}}$ (4) $2\sqrt{gC}$

CR0103

7. If $\vec{P} \times \vec{Q} = \vec{Q} \times \vec{P}$, the angle between $\vec{P}$ and $\vec{Q}$ is $\theta(0° < \theta < 360°)$. The value of '$\theta$' will be _____°.

[JEE Main-2021_Feb.]

CR0104

8. A block of 200 g mass moves with a uniform speed in a horizontal circular groove, with vertical side walls of radius 20 cm. If the block takes 40 s to complete one round, the normal force by the side walls of the groove is : **[JEE Main-2021_March]**

(1) 0.0314 N (2) 9.859×10^{-2} N (3) 6.28×10^{-3} N (4) 9.859×10^{-4} N

CR0105

9. **Statement I :** A cyclist is moving on an unbanked road with a speed of 7 kmh^{-1} and takes a sharp circular turn along a path of radius of 2m without reducing the speed. The static friction coefficient is 0.2. The cyclist will not slip and pass the curve ($g = 9.8\ m/s^2$)

Statement II : If the road is banked at an angle of 45°, cyclist can cross the curve of 2m radius with the speed of 18.5 kmh^{-1} without slipping. **[JEE Main-2021_March]**

In the light of the above statements, choose the correct answer from the options given below.

(1) Statement I is incorrect and statement II is correct

(2) Statement I is correct and statement II is incorrect

(3) Both statement I and statement II are false

(4) Both statement I and statement II are true **CR0106**

10. The angular speed of truck wheel is increased from 900 rpm to 2460 rpm in 26 seconds. The number of revolutions by the truck engine during this time is _____. (Assuming the acceleration to be uniform). **[JEE Main-2021_March]**

CR0107

11. The normal reaction 'N' for a vehicle of 800 kg mass, negotiating a turn on a 30° banked road at maximum possible speed without skidding is _____ $\times 10^3$ kg m/s^2. **[JEE Main-2021_July]**

(1) 10.2 (2) 7.2 (3) 12.4 (4) 6.96

CR0108

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12. A particle of mass m is suspended from a ceiling through a string of length L. The particle moves in a horizontal circle of radius r such that $r = \frac{L}{\sqrt{2}}$. The speed of particle will be :**[JEE Main-2021_Aug.]**

(1) $\sqrt{rg}$ (2) $\sqrt{2rg}$ (3) $2\sqrt{rg}$ (4) $\sqrt{\frac{rg}{2}}$

CR0109

13. For a particle in uniform circular motion, the acceleration $\vec{a}$ at any point P (R, θ) on the circular path of radius R is (when θ is measured from the positive x-axis and v is uniform speed) :

[JEE Main-2022_June]

(1) $-\frac{v^2}{R}\sin\theta\hat{i}+\frac{v^2}{R}\cos\theta\hat{j}$ (2) $-\frac{v^2}{R}\cos\theta\hat{i}+\frac{v^2}{R}\sin\theta\hat{j}$

(3) $-\frac{v^2}{R}\cos\theta\hat{i}-\frac{v^2}{R}\sin\theta\hat{j}$ (4) $-\frac{v^2}{R}\hat{i}+\frac{v^2}{R}\hat{j}$

CR0141

14. One end of a massless spring of spring constant k and natural length ℓ_0 is fixed while the other end is connected to a small object of mass m lying on a frictionless table. The spring remains horizontal on the table. If the object is made to rotate at an angular velocity ω about an axis passing through fixed end, then the elongation of the spring will be: **[JEE Main-2022_June]**

(1) $\frac{k-m\omega^2 l_0}{m\omega^2}$ (2) $\frac{m\omega^2 l_0}{k+m\omega^2}$ (3) $\frac{m\omega^2 l_0}{k-m\omega^2}$ (4) $\frac{k+m\omega^2 l_0}{m\omega^2}$

CR0142

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EXERCISE (J-A)

Paragraph for Question No. 1 and 2

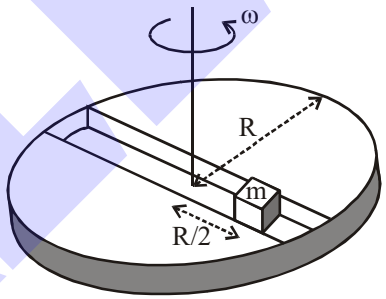
A frame of reference that is accelerated with respect to an inertial frame of reference is called a non-inertial frame of reference. A coordinate system fixed on a circular disc rotating about a fixed axis with a constant angular velocity ω is an example of a non-inertial frame of reference. The relationship between the force $\vec{F}_{rot}$ experienced by a particle of mass m moving on the rotating disc and the force $\vec{F}_{in}$ experienced by the particle in an inertial frame of reference is

$$\vec{F}_{rot} = \vec{F}_{in} + 2m(\vec{v}_{rot} \times \vec{\omega}) + m(\vec{\omega} \times \vec{r}) \times \vec{\omega},$$

where $\vec{v}_{rot}$ is the velocity of the particle in the rotating frame of reference and $\vec{r}$ is the position vector of the particle with respect to the centre of the disc.

Now consider a smooth slot along a diameter of a disc of radius R rotating counter-clockwise with a constant angular speed ω about its vertical axis through its center. We assign a coordinate system with the origin at the centre of the disc, the x-axis along the slot, the y-axis perpendicular to the slot and the z-axis along the rotation axis $(\vec{\omega} = \omega\hat{k})$. A small block of mass m is gently placed in the slot at $\vec{r} = (R/2)\hat{i}$ at t = 0 and is constrained to move only along the slot.

[IIT-JEE Advanced-2016]



1. The distance r of the block at time t is :

(A) $\frac{R}{4}(e^{2\omega t} + e^{-2\omega t})$ (B) $\frac{R}{2}\cos 2\omega t$ (C) $\frac{R}{2}\cos \omega t$ (D) $\frac{R}{4}(e^{\omega t} + e^{-\omega t})$

CR0078

2. The net reaction of the disc on the block is :

(A) $-m\omega^2 R\cos\omega t\hat{j} - mg\hat{k}$ (B) $m\omega^2 R\sin\omega t\hat{j} - mg\hat{k}$

(C) $\frac{1}{2}m\omega^2 R(e^{\omega t} - e^{-\omega t})\hat{j} + mg\hat{k}$ (D) $\frac{1}{2}m\omega^2 R(e^{2\omega t} - e^{-2\omega t})\hat{j} + mg\hat{k}$

CR0078

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ANSWER KEY

SUPPLEMENT EXERCISE

1. (D) **2.** $-10\hat{i}-7\hat{j}+4\hat{k}$, $\sqrt{165}$ **3.(B)** **4.** $18\hat{i}+4\hat{j}+6\hat{k}$

5. $15\hat{i}+6\hat{k}$ **6.(i)** ↓ ; **(ii)** ↑; **(iii)** ⊙; **(iv)** ← **7.** $-104\hat{k}$ Nm

8. (i) $-40\hat{k}$ **kgm²/s ; (ii)** $80\hat{k}$ **kgm²/s** **9.** ← **10. (C)** **11. (i) zero, (ii) yes**

12. (A) **13.** $\hat{n}=\frac{1}{\sqrt{42}}(-4\hat{i}+\hat{j}+5\hat{k})$ **14. (D)** **15. (A)**

17. z-direction, v = E/B **18.** $\frac{mg}{A}$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	B	C	D	A	C	D	C	
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	B	B	C	B	D	C	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	B	B	B	A	A	A	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	C	D	A,B	A,B	C,D	B,C,D	A,B,C,D	B,C	A
Que.	41	42	43	44	45					
Ans.	C	C	D	A	A-P,Q; B-P,Q,S; C-P,Q,R; D-P,Q,R					

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	B	D	B	B,D	A	B	A	C
Que.	11	12	13	14						
Ans.	D	B	C	A						

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EXERCISE # S

1. 9.9 ms^{-2}, in radial direction towards the centre at all points.

2. (i) 2 rad/s^2; **(ii)** 12 + 2t for $t \leq 2s$, 16 for $t \geq 2s$; **(iii)** a = 169.01 m/s^2 & 256 m/s^2 ; **(iv)** 44 rad

3. (i) 75m/s^2, **(ii)** 125m/s^2

4. 4 rad/s **5.** $\frac{10}{3}$ rad / s **6. 4** **7. 024** **8.** $\sqrt{3}b/2, b/2, b$

9. $\frac{v}{2d}$ **10.** $a_t = \frac{20}{\sqrt{13}}$, $a_n = \frac{30}{\sqrt{13}}$ **11.** (i) mg, (ii) $\frac{mv^2}{r}$ **12. 5**

13. $\sqrt{2g}$ rad/s **14.** T = 6.6 N, v_{max} = 35 ms^{-1} **15. 15 km** **16. 2**

17. (5 : 3) **18. Between 14.7 km/h and 54 km/hr.**

20. $\left[\frac{g(\sin\theta-\mu\cos\theta)}{R\sin\theta(\cos\theta+\mu\sin\theta)}\right]^{1/2}$ to $\left[\frac{g(\sin\theta+\mu\cos\theta)}{R\sin\theta(\cos\theta-\mu\sin\theta)}\right]^{1/2}$

21. (a) $\frac{mv^2}{R}$; (b) $\frac{\mu mv^2}{R}$; (c) $-\frac{\mu v^2}{R}$; (d) $v_0e^{-2\pi\mu}$ **22.** $\sqrt{\frac{2L}{\omega^2R\cos\theta}}$ **23. (a) 0.2 N; (b) 30°**

EXERCISE # JM

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	2	3	2	4	2	180	4	4	728
Que.	11	12	13	14						
Ans.	1	1	3	3						

EXERCISE # JA

Que.	1	2
Ans.	D	C

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